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GenAI: a scientometric analysis of research trends using biblioshiny and VOSviewer

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Abstract

Purpose Generative Artificial Intelligence (GenAI) has rapidly emerged as a transformative technology shaping diverse domains such as education, defense, healthcare, industry, tourism, and knowledge management. Despite its expanding influence, the scholarly landscape of GenAI research remains fragmented, making it challenging to trace its growth patterns, collaboration networks, and thematic evolution. To address this gap, this study conducts a comprehensive scientometric analysis to systematically map the intellectual structure and research dynamics of the field. The objective is to identify the most influential authors, institutions, countries, and thematic clusters, thereby providing a holistic view of the development of GenAI research.

Design/methodology/approach The study analyzes 9612 documents retrieved from Scopus and Web of Science databases (1986–2025, based on data availability). Data screening followed the PRISMA protocol to ensure transparency and rigor. Advanced bibliometric tools viz. Biblioshiny (R-based Bibliometrix) and VOSviewer were employed to examine productivity indicators, visualize co-authorship networks, and uncover thematic structures using keyword co-occurrence, trend analysis, and Latent Dirichlet Allocation (LDA) topic modeling.

Findings The results reveal a steep escalation in GenAI-related publications, increasing by 3164% in Period 2 (2022–24) compared to Period 1 (2012–22), reflecting the disruptive rise of text-generation tools such as OpenAI's GPT series, ChatGPT, and Gemini. The United States, China, India, and the United Kingdom dominate research output, with Harvard University, Nanyang Technological University, and the University of California leading institutionally. LDA topic modeling identifies ten major thematic clusters, spanning technical design, image synthesis, and large language models to applications in education, healthcare, legal governance, and ethical inquiry. Topic prevalence highlights governance, ethics, and risk (20%) along with AI-driven content creation (17%) as the most explored themes, followed by educational applications (14%). Lotka's Law analysis further reflects the characteristics of an emerging domain, with numerous occasional contributors and a smaller, prolific core group driving the research front.

Originality This study represents a comprehensive scientometric investigations of GenAI to date, integrating performance metrics, network visualizations, and advanced topic modeling. By mapping the intellectual structure, global collaborations, and

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evolving research trajectories, it offers actionable insights for scholars, practitioners, and policymakers seeking to navigate and contribute to this rapidly advancing field. The analysis reveals two dominant directions in GenAI research: governance across sectors and technical or application-specific innovation.

Keywords Generative artificial intelligence (GenAI), Scientometric analysis, Biblioshiny, VOSviewer, Latent dirichlet allocation (LDA)

1 Introduction

The evolution of Generative Artificial Intelligence (GenAI) can be traced through several distinct phases in the history of artificial intelligence. The earliest concepts emerged in the 1950 s, when Alan Turing introduced the idea of machine intelligence [1] and Joseph Weizenbaum developed ELIZA (1966), an early natural language program simulating conversation [2]. Although rule-based and limited, ELIZA demonstrated the potential of machines to generate human-like responses. The second phase, spanning the 1990 s to 2000 s, witnessed advances in statistical learning and neural networks, laying the groundwork for generative approaches [3]. The breakthrough moment came in 2014, when Ian Goodfellow and colleagues proposed the Generative Adversarial Network (GAN), a framework capable of generating highly realistic images and synthetic data [4]. GANs are widely regarded as the first modern GenAI tool. Parallel to progress in vision models, generative language models emerged with the introduction of the transformer architecture [5]. Building on this, OpenAI's Generative Pre-trained Transformer (GPT) series revolutionized natural language generation, beginning with GPT [6], followed by GPT-2 [7] and GPT-3 [8], which demonstrated unprecedented fluency and contextual understanding. The latest phase, post-2022, marks the mainstreaming of GenAI with tools such as ChatGPT, DALL·E, Stable Diffusion, and Google's Gemini, which have transformed creative industries, education, research, and knowledge management [9–11]. This rapid growth has led to an exponential increase in scholarly output, making scientometric analysis essential to map the trends, collaborations, and knowledge structures within GenAI research.

Artificial Intelligence (AI) has truly exploded over the last ten years, becoming a game-changer for education, defense, healthcare, industry, tourism and knowledge management. This substantial revolution has been catalyzed by advancements in neural networks and deep learning, particularly the transformer architecture [5], which has supplanted traditional methodologies for information processing in favor of "attention" mechanisms, that demonstrated remarkable efficacy in the comprehension of both linguistic constructs and visual representations. This transformer model is the foundation of all modern large language models (LLMs), and due to enhancements in training datasets and computational resources, these LLMs have acquired new competencies like remarkable speed and prompt response while requiring only minimal inputs [8].

The latest GenAI tools like ChatGPT and Google's Gemini are even more advanced. It is no longer just about text; these systems can now understand pictures, sound, and video, and even connect the dots between all of them at the same time. Collectively, these trajectories have fostered the accelerated growth of a distinct subfield, namely Generative AI systems, which are proficient in the creation of text, imagery, code, and other content that is of exceptionally high quality and controllability. GenAI offers a wide variety of model architectures (including autoregressive transformers, diffusion models,

and hybrid pipelines) that are proficient at producing original content when prompted in a precise manner. For instance, systems like DALL·E 2 seamlessly blend contrastive representation learning with generative decoders, enabling the transformation of linguistic inputs into remarkable visual representations [12]. Moreover, diffusion models have been innovatively designed to function by learning to eliminate noise from images, thus providing an efficient methodology for generating high-quality visual content [9]. In the realm of linguistic processing, instruction-tuned LLMs have attained mainstream acceptance through various chat interfaces (such as ChatGPT, Claude, Gemini, among others) and are now widely used in numerous sectors -research, software development, education, healthcare, law, to name a few. For example, in education field, these tools offer incredible features for personalized tutoring, instant feedback, and writing assistance but at the same time there are concerns to maintain academic integrity [13].

Businesses are experiencing enhancements in productivity, improved decision-making support, and novel modalities for value creation [14, 15] although they are simultaneously concerned about governance issues, potential workforce displacement, and the management of risks associated with these powerful technological tools. Healthcare professionals are utlizing Gen AI for generating medical summaries, diagnostic reports, and creating synthetic data for research [16, 17], though there are concerns about patient safety, reliability, privacy and ethical implications. Creative industries highlight that GenAI can help individual creators produce higher-quality work faster (Doshi & Hauser, 2024; Zhou & Lee, 2024). Yet there are growing concerns about whether this might lead to more homogenized content and questions the definition of original creativity.

As Generative AI has emerged as a transformative research domain in recent years, the growing body of Artificial Intelligence (AI) literature reflects a sustained interest in its evolution, applications, and ethical implications. Prior bibliometric and scientometric studies have explored AI research trends in various domains. For instance, Zhou et al. [18] conducted a scientometric study on global AI research output, identifying dominant countries and collaborative patterns. Similarly, Biswal et al. [19] analyzed machine learning and deep learning research, emphasizing the rapid expansion of transformer-based models and neural architectures. Other studies, such as Maheshwari et al. [20] and Jena et al. [21], have focused on the intersection of AI with innovation, financial technology [22], and behavioral sciences, using Biblioshiny and VOSviewer as core analytical tools. However, most of these studies adopt a broad focus on Artificial Intelligence and Machine Learning. Existing reviews tend to concentrate on technical developments or general trends, with limited attention to the scientometric mapping of GenAI's intellectual structure, thematic clusters, and collaborative dynamics [13, 15–17, 23, 24]. While there have been plenty of individual studies and reviews, the research landscape around GenAI still remains diffused. Early attempts to map this field have focused on a few specific domains like education, defense, healthcare, industry, tourism and knowledge management. Most of the studies did not address the cross-disciplinary structure of GenAI research, the evolution of author and institution networks; and the trajectory of methodological and thematic cores [23, 25–27]. Furthermore, the recent proliferation of GenAI tools, particularly since 2022, has introduced new research directions and ethical considerations that earlier bibliometric analyses did not capture. Therefore, a clear research gap exists in understanding how Generative AI research has evolved, who the key contributors are, and what thematic and geographical patterns define its growth.

While prior studies provided foundational insights into AI's global research landscape, none have comprehensively mapped GenAI-specific literature using combined Scopus and Web of Science datasets with both bibliometric and topic modeling approaches. Addressing this gap, the present study aims to offer a detailed scientometric mapping of Generative AI research covering the period from 1986 to 2025.

Scientometric studies are crucial because they provide systematic, evidence-based insights into the development of scientific fields. Unlike narrative reviews, scientometric approaches are data-driven and enable researchers to quantify productivity, identify influential authors and institutions, and map patterns of collaboration and knowledge diffusion [28, 29]. By applying bibliometric performance indicators and science-mapping techniques, scientometric analysis not only reveals the intellectual structure of a research domain but also highlights emerging themes and future directions. In the context of Generative Artificial Intelligence (GenAI), such an approach is particularly valuable given the field's rapid growth, interdisciplinary nature, and transformative potential across multiple sectors.

2 Literature review

Generative Artificial Intelligence (GenAI), has rapidly emerged as a paradigm-shifting technological force ever since launch of ChatGPT in November 2022. Globally, the professional landscape across diverse sectors are being reshaped by its adoption [30–32]. This advanced technology has impacted education field significantly by facilitating smart teaching and learning through personalised education, content creation, and adaptive training. This has underscored the transformative potential of GenAI tools in pedagogical contexts. The research community too has demonstrated an increasing interest in GenAI. There has been a surge in academic literature all across the globe and the same is also validated by one research paper having reported a notable 3,164% increase in GenAI publications. A steady increase in publications since 2018 has been observed and most notably an exponential growth during 2023–2024 has been witnessed [33–40].

This growing body of scholarly work is closely tied to the commercial success and widespread public adoption of powerful GenAI tools such as ChatGPT, which have acted as significant drivers for its acceptance [14, 36, 38, 41, 42]. With research growing at a fast pace, we need strong scientometric studies to understand the emerging trends and guide the future investigations [14, 33, 36, 43–46].

Evolving foundations of generative AI

Conceptually, the domain of Generative Artificial Intelligence (GenAI) is intrinsically interdisciplinary, integrating technological, ethical, and societal dimensions [38, 47, 48]. Although any model that generates output can be broadly categorized as generative, the AI research community frequently reserves the designation for sophisticated models that produce high-quality, human-like content, resulting in a significant divergence in comprehension between the general public and academic researchers [14]. The foundational principles that underpin these sophisticated systems consists of autoregressive models, Variational Autoencoders (VAEs), and Generative Adversarial Networks (GANs) [49]. In the beginning, research on Generative AI was all about the technology itself. Scientists were mainly focused on "deep neural networks" and the powerful ways they could be used to help computers see and understand images [35]. However, the research landscape is progressively evolving, with an increasing focus on explaining the

profound human and organizational implications of GenAI, including its extraordinary linguistic capabilities that frequently display human-like characteristics and prompt tendencies toward personification [35, 50].

Areas of applications of generative AI

A meticulous analysis of the academic literature unveils a rich and heterogeneous thematic landscape propelled by the versatile applications of GenAI. Education has become a huge focus for researchers, who are now looking closely at how this technology could completely change the way we teach and learn [31, 33, 34, 40, 44, 51]. Key applications currently under scrutiny encompass personalized learning trajectories, automated evaluation, chatbot-assisted instruction, as well as the facilitation of content generation and adaptive training methodologies [52, 53]. This enthusiasm is, however, tempered by substantial apprehensions regarding academic integrity, plagiarism, data security, and the potential degradation of cognitive abilities due to excessive dependence on artificial intelligence [33, 40, 54, 55].

In the domains of healthcare and medicine, applications of GenAI are focussed on the enhancement of patient care, clinical decision support, medical education, and therapeutic personalization, and this domain has witnessed a marked increase in scholarly publications observed since 2023 [56, 57]. In a similar vein, Marketing harnesses GenAI for the creation of dynamic content and personalised consumer interactions. Conversely, the tourism and hospitality sectors employ GenAI for decision-making processes and chatbot implementation, thereby emphasizing improved user experience and operational efficacy [58]. In Software Engineering domain, active research is evident in areas like code generation and debugging [59]. Across all sectors, ethical considerations and governance represent a pivotal, cross-cutting theme that pervades nearly all applications [30, 60]. Data privacy, algorithmic bias, misinformation, intellectual property rights, and the responsible utilization of artificial intelligence remain the most critical concerns [32, 61]. Additionally, there exist pressing security vulnerabilities inherent within these technologically advanced and complex models [54].

Methodological approaches in GenAI research

Methodologically, the domain of Generative Artificial Intelligence (GenAI) significantly depends on bibliometric analysis to articulate its intellectual framework and investigate research trajectories [38, 39]. VOSviewer constitutes a widely utilized software for the generation of network visualizations pertaining to keywords and co-citation patterns, thereby facilitating the identification of research focal points and thematic clusters [14, 38, 43, 44, 46, 47, 53, 57, 62–65]. In a similar vein, Biblioshiny, frequently employed with RStudio, is extensively leveraged for the analysis of data and the visualization of publication trends [38, 62, 65, 66].

The primary data repositories utilized in the research studies are predominantly the Web of Science (WoS) and Scopus databases [34, 35, 37, 38, 40, 44, 46, 47, 53, 54, 57, 61, 63, 64, 66–70]. Some studies augment their datasets by incorporating PubMed or Google Scholar searches' results [32, 34, 35, 42, 44, 46, 47, 53, 54, 57, 61, 63, 64, 66–72] Analytical techniques including keyword co-occurrence and co-citation analysis are prevalent for extractng thematic structures [53, 73], while techniques such as topic modeling and sentiment analysis yield more profound insights into research themes and community attitudes [32, 42, 54, 66]. The leading nations contributing to the field of GenAI publications are the United States, China, and the United Kingdom [40, 57, 58].

Gaps & objectives of the research

While recent advancements such as transformer architectures and large-scale models like ChatGPT have been widely recognized for their technological significance, most previous studies have examined them from algorithmic or application-oriented perspectives. Existing scientometric studies on Artificial Intelligence [18, 19] have focused mainly on general AI trends, without delving deeper into Generative AI's distinctive features creative content generation, transformer-based architectures, and ethical governance debates. While a few recent works have analyzed ChatGPT-related research growth, a large-scale analysis combining LDA topic modeling and scientometric mapping to capture the broader Generative AI landscape using integrated Scopus and Web of Science datasets is still lacking.

To address this gap, the present study provides a holistic scientometric overview of Generative AI research from 1986 to 2025. By combining traditional scientometric indicators with Latent Dirichlet Allocation (LDA) topic modeling, the analysis captures both structural and semantic dimensions of the field. This integrated approach enables a deeper understanding of how GenAI scholarship has evolved beyond technical advancements to encompass ethical, creative, and educational dimensions. Furthermore, the study employs keyword co-occurence, and network visualization techniques using Biblioshiny, Python, and VOSviewer to uncover the intellectual foundations and thematic evolution of the field.

Therefore, this multidimensional framework provides a comprehensive picture of global GenAI research trends. The findings not only highlight the surge in publications after 2022 and the widespread adoption of GenAI technologies but also reveal ongoing challenges that shape future innovation. The outcomes of this study aim to assist the research community, practitioners, and policymakers in fostering a nuanced understanding of emerging themes and collaborative dynamics within Generative AI research. Based on the identified research gap, the objectives of this study are:

(1) To analyze the publication growth and citation patterns of Generative AI research.
(2) To identify leading countries, authors & university/institutions and source.
(3) To visualize collaboration networks between countries and thematic structures.
(4) To analyze the hierarchical clustering of keywords (Dendrogram Analysis) and factorial analysis of Conceptual Structure Map.
(5) To explore thematic evolution through Latent Dirichlet Allocation (LDA) topic modeling.

3 Methodology

The present study employs a Scientometric analysis which enables quantitative evaluation of scientific publications to identify research trends, influential authors, collaboration networks, and emerging themes [28].

3.1 Formulation of search string

The keywords were identified through a review of prior scientometric studies on artificial intelligence and generative technologies [37, 57, 61, 74] which employed similar terminologies in bibliometric retrieval. The initial set of terms was pilot-tested in both Scopus and WoS databases to assess relevance and coverage. Subject experts in AI and bibliometrics validated the final search string to ensure comprehensive inclusion of

GenAI-related publications. The final search string combined core terms such as *genai OR "Generative AI" OR "gen ai" OR "AI-generated" OR "AI generated" OR "generat* AI" or "Generative artificial intelligence".* Boolean operators (OR) was employed to refine the scope and ensure the inclusion of all relevant records while avoiding unrelated studies.

3.2 Execution of search query

The formulated search string was executed on the Scopus and WoS databases because of their comprehensive metadata quality and citation reliability, and considering its wide coverage, reliable metadata, and citation indexing quality. The time span was restricted to 1986–2025 (since inception), ensuring both historical and contemporary research were captured. Only final records in the publication stage were included to provide complete bibliographic and citation information, while restricting the dataset to English-language publications ensured data reliability and reflected the dominant language of global AI research. After conducting a search using carefully selected and validated keywords [75, 76], the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [77, 78] framework was employed to ensure transparency and reproducibility of the exported data. Figure 1 depicts the PRISMA process which consists of: identification, screening, and inclusion. The query execution resulted in the initial retrieval of 13,958 documents (from Scopus & WoS databases). The search was conducted on 26th Aug 2025, using the following search strings:

Scopus: TITLE (genai OR "Generative AI" OR "gen ai" OR "AI-generated" OR "AI generated" OR "generat* AI" OR "Generative artificial intelligence") AND (EXCLUDE (PUBYEAR, 2026)) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (PUBSTAGE, "final")).

WoS: genai OR "Generative AI" OR "gen ai" OR "AI-generated" OR "AI generated" OR "generat* AI" or "Generative artificial intelligence" (Title) and 2026 (Exclude – Publication Years) and English (Languages) and 2026 (Exclude – Final Publication Year) and 2025 or 2024 or 2023 or 2022 or 2021 or 2020 or 2019 or 2017 (Final Publication Year).

3.3 Data collection and data cleaning

Following data retrieval, a rigorous data cleaning was undertaken to ensure data quality and reliability. The raw data retrieved from Scopus and Web of Science (WoS) were first imported into the Biblioshiny software (a web application of the bibliometrix R package - version 5.1.1) for initial processing. Since, the WoS database allows downloading in batches of 500 records only, so merging of seven raw WoS files was executed that resulted in a single file for 3022 WoS records in excel format. Similarly, the Scopus database file having 9285 records in csv was converted to excel format using biblioshiny. Following this, the merging of Scopus and WoS was performed using the function mergeDbSources(Scopus, WoS, remove.duplicated = TRUE) available in RStudio (version 4.5.1) that matches key bibliographic identifiers such as DOI (Digital Object Identifier) and title, and drops one of the duplicate entries, keeping the most complete one (the one with more filled fields). This function identified 2681 records (out of 12,307) as duplicate and hence were dropped which resulted in 9626 unique records.

Data cleaning was carried out based on standardized field headings generated by Biblioshiny. Records lacking essential bibliographic information particularly those without author details were manually identified and hence 14 records such incomplete records

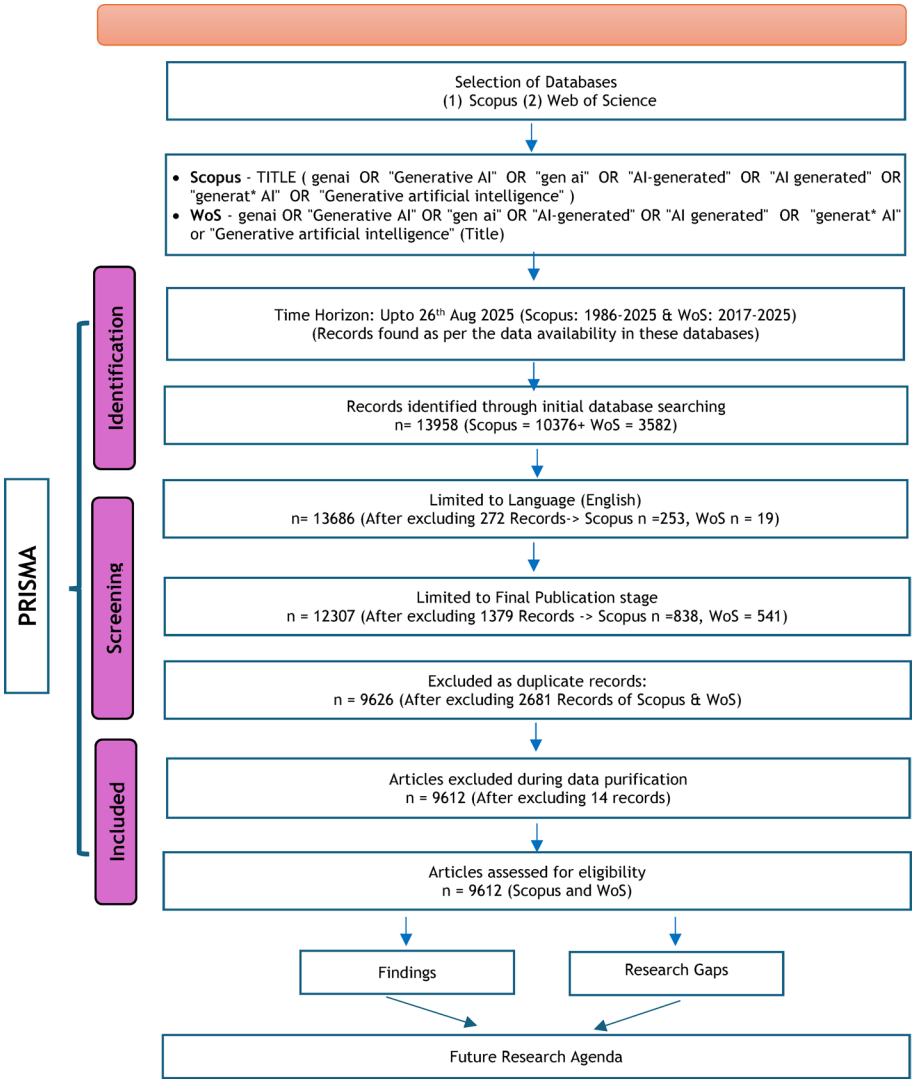

Fig. 1 PRISMA approach showing Data Capturing & Filtration

were removed. The final merged comprised 9612 valid records that were used for all scientometric analysis.

3.4 Application of scientometric tools and technique

The processed dataset was analysed using advanced scientometric tools such as Biblioshiny [77] and VOSviewer [79]. Biblioshiny was employed to generate descriptive bibliometric indicators including publication growth trends, authorship productivity, leading sources, and citation impact. VOSviewer was applied for mapping and visualization, enabling the construction of co-authorship networks between countries to find out country-level linkages. The combined use of these methods ensured a robust and comprehensive scientometric analysis of the research field. Latent Dirichlet Allocation (LDA) topic modeling was also applied to explore thematic structures and emerging research directions in GenAI on the basis of abstracts of research papers. The LDA model was trained using following function in Jupyter Notebook (Version: 7.0.8) for Python: -.
LdaModel(

```
  corpus = corpus,
  id2word = dictionary,
  num_topics = 10, # Adjust topics here.
  random_state = 42,
  passes = 10,
  alpha='auto'
)
```

where, corpus corresponds to cleaned abstracts (obtained after excluding stopwords, performing stemming and lemmatization). The parameter id2word = dictionary links each word's unique ID in the corpus to its actual text, enabling the LDA model to display interpretable word-based topics instead of numeric tokens. The parameter num_topics = 10 specifies extraction of ten latent themes; passes = 10 controls how many times the model iterates over the data for better convergence; alpha='auto' lets the model learn optimal topic–document distribution; and random_state = 42 ensures reproducible results.

4 Finding

4.1 Growth of publications

The bibliometric analysis in Table 1 examines a collection of 9612 documents published between 1986 and 2025, sourced from 3528 distinct sources (journals, books, etc.). The relatively recent timespan of the data, coupled with a high annual growth rate of 23.86%, suggests a rapidly evolving and expanding field of study. The document average age of only 0.719 years further reinforces this notion of a current and dynamic area of research. The substantial number of documents authored by 23,002 authors and the relatively high co-authors per document ratio of 3.78 indicate a collaborative research environment. While 1659 documents are single-authored, highlighting a degree of individual contribution, the overall trend leans towards collaborative projects. The international co-authorship percentage of 9.863% suggests a moderate level of global collaboration, which could be further investigated to identify key international partners and collaboration networks.

Table 1 Annual publications on GenAI research depicting No. of publications (NP) and total citations (TC) (1986–2025)

Year	NP	TC	Mean TC per article	Mean TC per year	Citable years
1986	1	1	1.00	0.03	40
1990	2	52	26.00	0.72	36
1991	1	11	11.00	0.31	35
2004	1	0	0.00	0.00	22
2007	1	2	2.00	0.11	19
2011	1	3	3.00	0.20	15
2013	1	32	32.00	2.46	13
2017	2	51	25.50	2.83	9
2018	2	4	2.00	0.25	8
2019	6	59	9.83	1.40	7
2020	21	581	27.67	4.61	6
2021	36	845	23.47	4.69	5
2022	59	1726	29.25	7.31	4
2023	948	25,643	27.05	9.02	3
2024	4315	26,494	6.14	3.07	2
2025	4215	5142	1.22	1.22	1
		60,647			

The distribution of document types provides insight into the nature of research communication within this field. Articles (3854) and conference papers (3459) constitute the bulk of the publications, indicating a strong emphasis on both formal journal publications and presentations at academic conferences. The presence of book Chaps. (885), books (113) and reviews (289) suggests the existence of established knowledge and attempts to provide synthesis and overview of the research. The average citation per document is 6.308, which might be considered low and might be affected by the fact that most publication years are near the present. It might also reveal information about the interdisciplinarity of the research area, the specific journals used for the analysis, the type of documents and a wide range of other factors. The presence of author's keywords (16,461) and keywords plus (20,505) indicates well-defined research topics and areas of interest within the collection. Further analysis of these keywords could reveal the main themes, emerging trends, and intellectual structure of the field. This bibliographic collection represents a rapidly growing, collaborative, and multifaceted area of research.

The annual growth of publications on Generative AI demonstrates a significant upward trend, particularly after 2021. Between 1986 and 2010, the number of publications remained relatively low and stable. However, starting from 2018, a sharp increase is observed, peaking in 2023 and 2024. This rise corresponds to the widespread academic and industrial adoption of large language models (LLMs) and transformer-based architectures. The cumulative citation pattern reflects a similar trajectory, indicating strong and sustained scholarly interest in this emerging field. Also, the recent publications especially from year 2023 onwards have limited citation windows, hence fewer citable years. Still, the average citation per document also increased over time, showing that more recent publications have gained considerable attention within a short period. This growth pattern underscores Generative AI as one of the fastest-growing research domains in artificial intelligence.

4.2 Geographical distribution and collaboration patterns

Table 2 presents the ten most productive countries in Generative AI research (1986–2025), ranked by publication count, total link strength (TLS), and citation impact. The United States leads decisively with 1770 publications and 13,466 citations, followed by India (917) and China (654). While India ranks high in output, its lower citation rate (2.8) suggests a growing yet less-cited body of work. China and the United Kingdom demonstrate both substantial productivity and stronger impact, with the U.K. and

Table 2 Top 10 cited countries based on no. of documents published in GenAI research (1986–2025)

Country	Documents (Total link strength)	Frequency of country	TC	Average Article citations
United States	1770 (896)	4219	13,466	8.40
India	917 (388)	1098	1376	2.80
China	654 (402)	2327	6316	5.60
United Kingdom	502 (585)	912	3140	8.90
Germany	362 (341)	594	2342	9.30
Australia	315 (314)	625	3022	11.60
Canada	207 (268)	379	700	5.30
Italy	206 (182)	425	840	4.80
Spain	166 (193)	300	608	4.70
South Korea	161 (117)	491	897	3.60

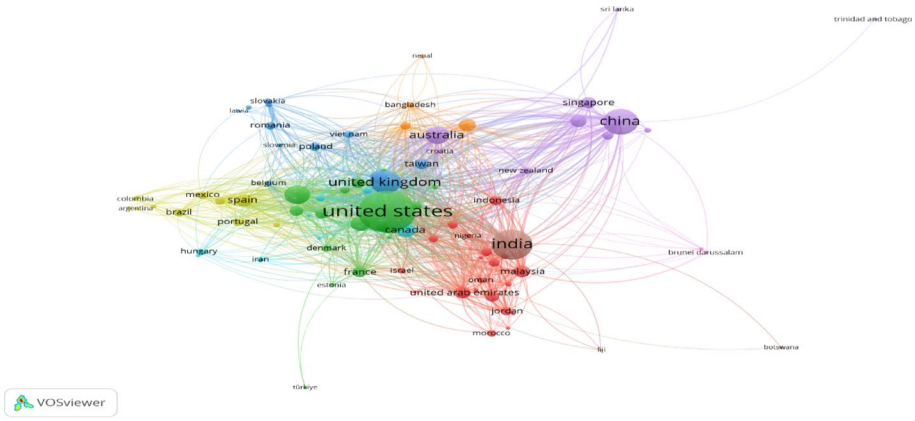

Fig. 2 Co-Authorship of Countries of GenAI Research using VOSviewer

Table 3 Top 10 most relevant affiliations with country based on no. of articles published

Affiliation	Country	Articles
Nanyang Technological University	Singapore	179
Harvard University	USA	178
University of California System	USA	164
Monash University	Australia	134
Zhejiang University	China	128
Shanghai Jiao Tong University	China	116
Stanford University	USA	113
Tsinghua University	China	109
Harvard University Medical Affiliates	USA	96
Carnegie Mellon University	USA	89

Australia showing the highest average citation rates (8.9 and 11.6), indicating emphasis on research quality and selective collaboration. Overall, GenAI research remains concentrated in technologically advanced economies, yet emerging nations—particularly India and China—are expanding their global presence.

The VOSviewer co-authorship visualization (Fig. 2) reveals three principal collaboration clusters centered on the United States, India, and China. The U.S. cluster exhibits dense transatlantic and Asia-Pacific linkages with the U.K., Germany, Australia, and Canada, reflecting a mature collaborative ecosystem. India's cluster connects with Southeast Asian and Middle Eastern countries, while China's cluster shows strong intra-Asian cooperation with Singapore and Taiwan. Peripheral yet emerging collaborations appear from nations such as South Korea, Italy, and Spain. These patterns indicate that while established economies continue to drive research output, developing regions are increasingly integrated into the global GenAI network, signaling a gradual shift toward a more diversified and interconnected research landscape. While citation averages differ across countries, it may be influenced by citable years and document type. Hence, the analysis focuses primarily on overall research output and collaborative trends rather than raw citation counts.

Table 3 highlights the most relevant affiliations contributing to GenAI research, ranked by the number of articles produced. Nanyang Technological University (Singapore) leads with 179 articles, closely followed by Harvard University (USA) with 178 articles, and the University of California System (USA) with 164 articles. These three

institutions account for a significant share of global GenAI research output. Institutional analysis highlighted that most high-impact contributions in terms of no. of articles published originated from globally recognized universities and research laboratories, including Nanyang Technological University, Harvard University, and University of California System. The high level of international collaboration suggests that Generative AI is an inherently interdisciplinary and globally shared area of investigation. Monash University (Australia) is a leading institution from the Asia-Pacific region, contributing 134 articles and indicating strong GenAI research activities outside the U.S. and China. Apart from Nanyang Technological University, Zhejiang University (China), Shanghai Jiao Tong University (China), and Tsinghua University (China) are notable Asian contributors, reflecting China's strong institutional research capacity in GenAI. The United States dominates the list with multiple top institutions – Harvard University, University of California System, Stanford University, Harvard University Medical Affiliates, and Carnegie Mellon University – underscoring its academic and research leadership. The institutional distribution shows that GenAI research is heavily concentrated in world-class universities and research centers in technologically advanced countries, primarily the United States, China, and Singapore. The presence of multiple U.S. institutions in the top rankings emphasizes the country's strong academic infrastructure and funding capabilities. Asian universities, particularly from China and Singapore, highlight the region's rapidly growing influence in cutting-edge AI research. Overall, the data reflects a competitive and globally distributed research landscape, with collaborative potential between leading institutions.

4.3 Three-field plot relationship

The three-field plot in Fig. 3 shows the relationships between author affiliations (AU_UN), author countries (AU_CO), and individual authors (AU). This visualization shows which institutions and countries are contributing the most to the dataset and how individual authors are distributed across them. The plot shows a network of connections, where the thickness of the connecting lines represents the strength or frequency of the relationship between the different fields.

Dominant Countries: The "AU_CO" field clearly indicates that the "USA" and "China" are the dominant countries in this dataset. The wider bands connecting them to the "AU_UN" and "AU" fields illustrate a substantial contribution to the global authorship. The dominance of the USA and China is evident. This indicate the leading role of these

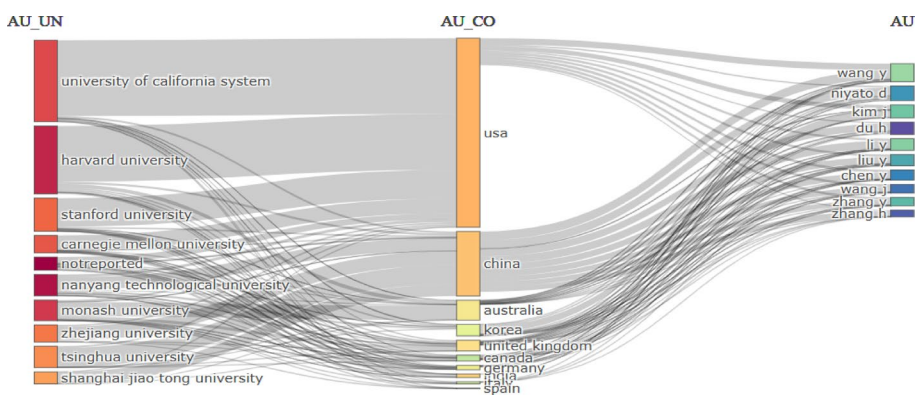

Fig. 3 Three-field Plot linking institutions, countries, and key authors in GenAI research

countries in the research area covered through bibliometric analysis. It would be beneficial to investigate the reasons for this prominence, considering factors like research funding, infrastructure, and national research priorities. The presence of Australia, Korea, and the UK indicates relevant contributions from other countries.

Top Affiliations: In the "AU_UN" field, the "University of California system", "Harvard University", and "Stanford University" are the most prominent affiliations, largely connected to the USA. This suggests a strong research output from these institutions. Following these, several Chinese Universities appears like "Tsinghua University, Zhejiang University, and Shanghai Jiao Tong University". The University of California System, Harvard, and Stanford appear as major nodes, suggesting that a significant portion of the publications and originates from these institutions. This reflect the research focus of these universities or their prominence in the relevant fields covered.

Author Distribution: The "AU" field shows a distribution of individual authors. Notice how certain authors, such as *Wang Y* and *Zhang Y/H*, have strong connections to both the USA and China through different affiliations, suggesting possible collaborations or authors moving between institutions. The spread of connections also suggests the degree to which these authors are working solely within one country or collaborating internationally. The distribution of individual authors and their connections to different countries and institutions provides insights into research collaborations and mobility. Authors connected to multiple countries might be involved in international collaborations, while those strongly linked to specific institutions likely conduct their research primarily within those institutions.

The connections between universities in different countries and individual authors suggest the presence of international collaboration. The prominence of specific universities like the University of California system, Harvard, and Stanford indicates their influence in the field.

The differences in productivity among countries also suggest variations in research funding and collaborative ecosystems. For instance, the United States and China host large public–private partnerships and national AI initiatives that provide strong institutional and financial support for GenAI research. This systemic backing not only increases publication volume but also enhances the visibility and citation potential of their work in global databases.

4.4 Most influential researchers

The citation analysis indicated that the most influential papers were those that introduced foundational generative models such as Generative Adversarial Networks [4] and Transformer architectures [5]. These works consistently appeared as top-cited references within the dataset. Authors associated with pioneering institutions in AI research exhibited high h-index values, reinforcing their substantial influence in shaping GenAI scholarship.

The Table 4 highlights the leading researchers contributing significantly to Generative AI (GenAI) research based on metrics such as h-index [80], g-index [81], total citations (TC), number of publications (NP), and highest citation impact. Niyato D (h-index 19, TC 995), Du H (h-index 17, TC 829), and Kang J (h-index 16, TC 763) emerge as the most impactful researchers, with their collaborative work on Edge-Cloud Generative AI in Mobile Networks (2024) being the most cited (108 citations). Wang Y stands out with

Table 4 Top 10 most productive authors ranked by h-index, calculated from publications within the generative AI corpus using biblioshiny

Author (Affiliation)	h-Index	g-Index	TC	NP	PY start	Highest TC of Best Paper (Yr)	Title of paper having highest citations (year)	Best paper published source	DOI
Niyato D	19	30	995	67	2023	108 (2024)	Unleashing the Power of Edge-Cloud Generative AI in Mobile Networks: a Survey of AIGC Services (2024)	IEEE Communications Surveys and Tutorials	https://doi.org/10.1109/Comst.2024.3353265
Du H	17	28	829	50	2023	108 (2024)	Unleashing the Power of Edge-Cloud Generative AI in Mobile Networks: a Survey of AIGC Services (2024)	IEEE Communications Surveys and Tutorials	https://doi.org/10.1109/Comst.2024.3353265
Kang J	16	27	763	44	2023	108 (2024)	Unleashing the Power of Edge-Cloud Generative AI in Mobile Networks: a Survey of AIGC Services (2024)	IEEE Communications Surveys and Tutorials	https://doi.org/10.1109/Comst.2024.3353265
Wang Y	15	27	805	100	2021	189 (2023)	A Survey on ChatGpt: AI-Generated Contents, Challenges, and Solutions (2023)	IEEE Open Journal of the Computer Society	https://doi.org/10.1109/Ojcs.2023.3300321
Xiong Z	15	23	570	34	2023	108	Unleashing the Power of Edge-Cloud Generative AI in Mobile Networks: a Survey of AIGC Services (2024)	IEEE Communications Surveys and Tutorials	https://doi.org/10.1109/Comst.2024.3353265
Kim J	13	26	737	79	2022	78	Generative Artificial Intelligence to Transform Inpatient Discharge Summaries to Patient-Friendly Language and Format (2024)	Jama Network Open	https://doi.org/10.1001/Jamanetworkopen.2024.0357
Kim D	12	21	447	30	2020	108	Unleashing the Power of Edge-Cloud Generative AI in Mobile Networks: a Survey of AIGC Services (2024)	IEEE Communications Surveys and Tutorials	https://doi.org/10.1109/Comst.2024.3353265
Li Y	12	19	421	81	2023	78 (2025)	Factors Influencing University Students' Behavioral Intention to use Generative Artificial Intelligence: Integrating the Theory of Planned Behavior and AI Literacy (2025)	International Journal of Human-Computer Interaction	https://doi.org/10.1080/10447318.2024.2383033
Liu Y	12	19	400	69	2022	131 (2023)	Generative Artificial Intelligence and its Applications In Materials Science: Current Situation and Future Perspectives (2023)	Journal of Materiomics	https://doi.org/10.1016/J.Jmat.2023.05.001
Zhang H	12	20	438	53	2023	83 (2024)	Modelling Generative AI Acceptance, Perceived Teachers' Enthusiasm And Self-Efficacy to English as a Foreign Language Learners' Well-Being in the Digital Era (2024)	European Journal of Education	https://doi.org/10.1111/Ejed.12770

the highest citation (189 in 2023) for a survey paper on ChatGPT and AI-generated content, reflecting strong academic influence in emerging AI discourse. Multiple authors (e.g., Niyato D, Du H, Kang J, Kim D, Xiong Z) share the same high-impact publication, indicating collaborative clusters in advancing mobile network applications of GenAI. IEEE Communications Surveys and Tutorials dominates as the preferred journal, signifying its prominence in disseminating high-impact GenAI research. High citation counts within a short span (2023–2025) suggest that GenAI research is rapidly evolving, with certain papers quickly becoming seminal references in the field.

The data reveals that a small group of prolific researchers drives a significant portion of impactful GenAI research, particularly in mobile networks, AI-generated content, and educational applications. High citation numbers within a short time frame reflect the growing relevance of GenAI and its interdisciplinary applications in both technology and human-computer interaction domains.

Lotka's law of scientific productivity

Lotka's Law in Table 5 evaluates the distribution of author productivity in GenAI research by analyzing how many authors contribute a specific number of publications. Out of 23,002 authors, 18,137 (78.8%) contributed only one paper, showing that the majority of contributors are occasional authors in the GenAI domain. The number of authors decreases sharply as the number of publications increases. Only 2,818 authors (12.3%) published two papers, and just a small fraction contributed three or more papers. A very small number (around 119 authors) produced between 1647 papers, indicating a core group of prolific contributors driving significant research output.

Observed values closely align with Lotka's theoretical predictions, which suggest that author productivity follows an inverse-square distribution - i.e., the number of authors publishing n papers is about $1/n^2$ of those publishing one paper.

The analysis confirms that GenAI research follows Lotka's Law, with a large base of occasional contributors and a small core of highly productive authors shaping the field. This pattern is typical of emerging and multidisciplinary research domains, where participation is widespread but sustained productivity is concentrated among a few leading scholars and research groups.

4.5 Source and article impact

The analysis of influential sources in Table 6 highlights the key journals and conference proceedings contributing significantly to the research field. Among the sources, "Computers and Education: Artificial Intelligence" and "Education Sciences" stand out with high citation counts despite being relatively recent (2023). This indicates a strong and immediate impact of published research, reflecting the growing relevance of AI and technology in education. Lecture Notes in Computer Science (LNCS), with publications dating back to 1990, shows consistent contributions but a comparatively moderate h-index and g-index, suggesting that while the volume of papers is high, citation influence per paper is distributed over a long period. Conference proceedings dominate the landscape in terms of total citations (TC) and number of publications (NP), underscoring their role as primary dissemination platforms for emerging research. The data confirms that a blend of high-impact journals and prolific conference proceedings drives scholarly influence in this domain, with recent journals achieving rapid recognition due to timely and innovative research topics.

Table 5 Authorship productivity distribution based on lotka's law of scientific productivity in GenAI research

Documents written	No. of authors	Proportion of authors	Theoretical
1	18,137	0.788	0.616
2	2818	0.123	0.154
3	869	0.038	0.068
4	365	0.016	0.038
5	226	0.01	0.025
6	130	0.006	0.017
7	91	0.004	0.013
8	86	0.004	0.01
9	39	0.002	0.008
10	40	0.002	0.006
11	24	0.001	0.005
12	27	0.001	0.004
13	15	0.001	0.004
14	16	0.001	0.003
1647 (vary from 15–100)	119	0.001	0.027
Total	23,002		

Table 6 Leading top 10 core journals based on h-index dominating research output in GenAI research

Source	Conference/Journal	h-index	g-index	TC	NP	PY start
Conference on Human Factors in Computing Systems - Proceedings	Conference Proceedings (CHI Conference, ACM SIGCHI)	22	38	1743	179	2020
Computers and Education: Artificial Intelligence	Journal (Elsevier)	19	40	1679	54	2023
ACM International Conference Proceeding Series	Conference Proceedings (ACM)	15	30	1127	149	2007
Education Sciences	Journal (MDPI)	13	34	1217	46	2023
Journal of Applied Learning and Teaching	Journal (Kaplan Business School, Australia)	11	18	363	25	2023
IEEE Access	Journal (IEEE open-access mega-journal)	10	24	640	81	2023
Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)	Conference Proceedings Series (Springer publishes LNCS volumes for many conferences, including AI & bioinformatics)	10	13	352	125	1990
Sustainability	Journal (MDPI)	10	25	629	25	2023
Education and Information Technologies	Journal (Springer)	9	18	382	37	2024
International Journal of Educational Technology in Higher Education	Journal (Springer)	9	21	1085	21	2023

The analysis of source journals indicated that the majority of publications on Generative AI appeared in journals focusing on computer science, engineering, information systems, and applied artificial intelligence. The results suggest that Generative AI research is expanding across disciplinary boundaries, reflecting its growing influence in diverse scientific domains. Subject area distribution confirmed that computer science contributed the highest proportion of publications, followed by engineering, decision sciences, and social sciences. This multidisciplinary spread indicates that Generative AI research

has evolved beyond its technical roots, finding applications in educational, creative, and ethical contexts.

Impact of most cited article

The Table 7 highlights the ten most cited documents in GenAI research, emphasizing their scholarly influence through total citations (TC), yearly citation rates (TC per Year), and normalized citation counts (Normalized TC).

- Top-Performing Article: Fui-Hoon N F (2023) in J Inf Technol Case Appl Res leads with 726 citations, the highest TC per Year (242.00) and Normalized TC (26.84). This suggests exceptional impact and early recognition within the research community.
- High-Impact Cluster: The next influential papers by Pavlik J (2023) and Chan C (2023) have 628 and 498 total citations, respectively, indicating a strong core of widely referenced work.
- Citation Density: The average TC per Year of most papers exceeds 140, showing rapid citation accumulation, a common trait in emerging, high-interest fields like GenAI.
- Journal Diversity: These papers are distributed across multiple journals (J Inf Technol Case Appl Res, Nature, Science, Educ Sci, etc.), reflecting interdisciplinary interest—spanning education, technology, science, and medical fields.

The dominance of 2023 publications suggests GenAI is a fast-growing research frontier, with new articles gaining rapid traction. High normalized citation values indicate that these studies not only receive widespread attention but also serve as foundational references shaping future research directions.

4.6 Thematic evolution

The strategic and thematic evolution analysis reveals how the intellectual landscape of Generative AI (GenAI) research has matured over nearly four decades (1986–2024 and 2025–2025), moving from foundational technical explorations to broader

Table 7 Top 10 most cited documents in GenAI research

Paper	DOI	Total citations	TC per year	Normalized TC
Fui-Hoon N F, 2023, J Inf Technol Case Appl Res	https://doi.org/10.1080/15228053.2023.2233814	726	242.00	26.84
Pavlik J, 2023, Journal Mass Commun Educ	https://doi.org/10.1177/10776958221149577	628	209.33	23.22
Chan C, 2023, Int J Educ Technol H	https://doi.org/10.1186/s41239-023-00411-8	498	166.00	18.41
Cooper G, 2023, J Sci Educ Technol	https://doi.org/10.1007/s10956-023-10039-y	494	164.67	18.26
Lim W, 2023, Int J Manag Educ-Oxf	https://doi.org/10.1016/j.ijme.2023.100790	485	161.67	17.93
Stokel-Walker C, 2023, Nature	https://doi.org/10.1038/d41586-023-00340-6	479	159.67	17.71
Noy S, 2023, Science	https://doi.org/10.1126/science.adh2586	445	148.33	16.45
Qadir J, 2023, Ieee Global Eng Edu Conf, Educon	https://doi.org/10.1109/EDUCON54358.2023.10125121	439	146.33	16.23
Meskó B, 2023, Npj Digit Med	https://doi.org/10.1038/s41746-023-00873-0	432	144.00	15.97
Michel-Villarreal R, 2023, Educ Sci	https://doi.org/10.3390/educsci13090856	419	139.67	15.49

interdisciplinary engagements. The thematic evolution in Fig. 4 reveals several interesting trends. The analysis was performed using the "DE" field (likely "Descriptors" or "Keywords Plus") with specific parameters ($n = 250$, n-grams = 1 to capture dominant thematic terms) to identify key themes.

Between 1986 and 2024, foundational themes such as deep learning and generative adversarial networks laid the groundwork for contemporary developments, gradually evolving into the broader and more integrative concepts of generative AI and generative artificial intelligence observed in 2025. The persistence of training as a recurring theme highlights its ongoing methodological relevance, and the higher education as a standalone topic has evolved into broader AI application contexts. Collectively, the map illustrates a clear thematic convergence, from fragmented subfields to a unified focus on generative models and their transformative, cross-sectoral potential, reflecting the field's progression from algorithmic exploration to applied, multidisciplinary innovation.

To ensure temporal relevance, the analysis distinguishes two distinct periods: the first (1986–2024) capturing the foundational and developmental trajectory of GenAI scholarship, and the second (2025–2025) deliberately included to represent the most recent wave of emerging themes based on publications available as of April 2025. Although this latter dataset is relatively limited in size, its inclusion offers critical insight into the field's current research directions and evolving intellectual frontiers.

4.7 Thematic map & keyword co-occurrences

Building on these evolutionary insights, the subsequent thematic evolution map (Fig. 5) provides a deeper understanding of how the identified themes interconnect within the broader intellectual structure of Generative AI research. Constructed from 250 keywords with the top 3 labels displayed for each cluster, the keyword co-occurrence map visualizes the major research concentrations shaping the field. It highlights four dominant thematic areas viz. deep learning, ethical AI, creative generation, and education-focused applications, each positioned according to its centrality (relevance) and density (development). This multidimensional mapping not only reveals the maturity and influence of core research domains but also illustrates the evolving balance between well-established areas and emerging frontiers within Generative AI scholarship.

The map is divided into four quadrants based on centrality (relevance) and density (development).

1. **Motor Themes (High Centrality, High Density)**: This quadrant contains the most influential and well-developed themes. We see two distinct clusters here:

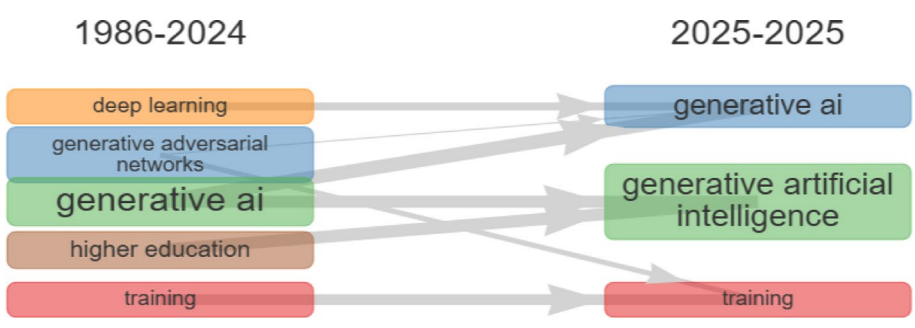

Fig. 4 Thematic Evolution of GenAI research from1986–2024 and 2025–2025

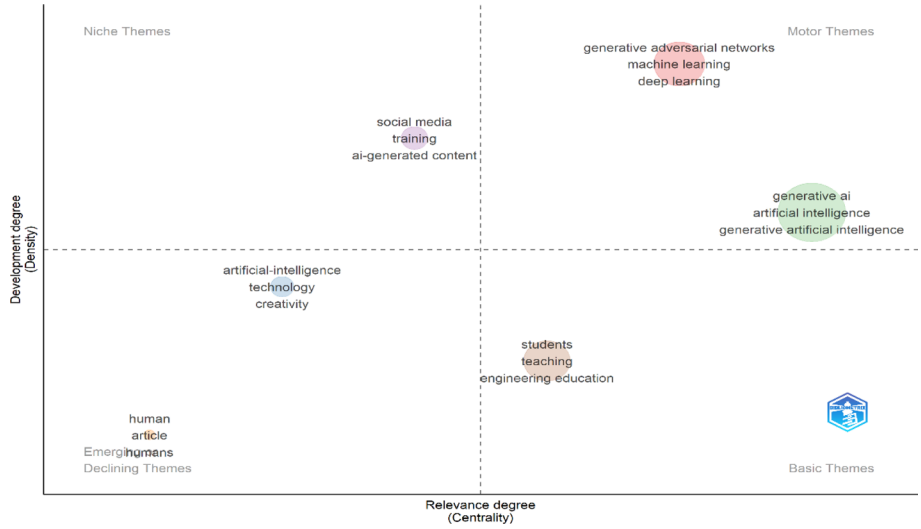

Fig. 5 Thematic Evolution Map showing Basic, Motor, Emerging and Niche Themes of GenAI research from 1986–2025

Generative AI and Artificial Intelligence: This cluster, represented by keywords like "generative AI," "artificial intelligence," and "generative artificial intelligence," is a core theme. The articles with the highest PageRank include Prather J, 2023, ITICSE-WGR - Proc Working Group Reports Innov Technol Comput Scie Educ (0.3), Kozhanov M, 2024, Cinti - IEEE Int Symp Comput Intell Informatics, Proc (0.297), and Aamir S, 2024, ISMSIT - Int Symp Multidiscip Stud Innov Technol, Proc (0.271). The high centrality and density suggest this area is both actively researched and strongly connected to other themes in the dataset. The presence of "generative ai" indicates a significant current interest and growth area.

Generative Adversarial Networks, Machine Learning, and Deep Learning: The location of this cluster in the motor themes quadrant indicates that it is a well-developed and central area of research. The most central articles include Celik A, 2024, IEEE Open J Commun Soc (0.287), Solanki S, 2024, Generative Artificial Intelligence: Exploring the Power and Potential of Generative Ai (0.249), and Cohan P, 2024, Brain Rush: How to Invest and Compete in the Real World of Generative Ai (0.244). This cluster is likely a fundamental driver in the broader AI landscape.

2. **Basic Themes (High Centrality, Low Density)**: This quadrant identifies themes that are central to the field but less developed or researched extensively in this dataset. The presence of "students," "teaching," and "engineering education" indicates a focus on the application of AI in education, particularly in engineering. The most central articles include Khatib R, 2024, Int Conf Inf Syst, ICIS (0.248), Jauhiainen J, 2025, IST-Africa (0.235), and Fan J, 2025, Proc Annu Hawaii Int Conf Syst Sci (0.234).
3. **Niche Themes (Low Centrality, High Density)**: These themes are specialized and well-developed but have less connection to the main research streams.

   Social Media, Training, and AI-Generated Content: This cluster suggests a focused interest in the use of AI for content creation and its implications for social media platforms. The most central articles include Zhang P, 2024, Future Internet (0.2), Górtowski S, 2025, Lect Notes Bus Inf Process (0.175), And Sai S, 2024, IEEE Access-A (0.174).

Artificial Intelligence, Technology, and Creativity: The placement of this cluster suggests that while AI is central to the area, its specific connections to 'technology' and 'creativity' are more niche. The most central articles include Wessel M, 2025, J Manage Inform Syst (0.17), Barus O, 2025, Int Conf E-Educ, E-Bus, E-Manag E-Learn, IC4E (0.165), and Huynh A, 2025, Lecture Notes Data Eng Commun Tech (0.151).

4. **Emerging or Declining Themes (Low Centrality, Low Density)**: As per our dataset, this quadrant contains themes appear as emerging or transitional rather than declining. Keywords such as artificial-intelligence, technology, and creativity indicate areas that are developing. Their placement reflects ongoing conceptual evolution, where these themes have not yet formed strong internal cohesion or strong connections with other clusters. The most central articles include Balsa-Barreiro J, 2024, Curr Top Behav Neurosci (0.169), Rosen J, 2024, Curr Gastroenterol Rep (0.165), And Currie G, 2025, J Nucl Med Technol-A (0.162). As research activity increases, these emerging themes may progress toward more central and mature thematic positions.

4.8 Hierarchical clustering of keywords (Dendrogram Analysis)

The dendrogram in Fig. 6 illustrates the hierarchical clustering of keywords associated with GenAI research, revealing thematic groupings and conceptual relationships. The dendrogram reveals three major research directions: "Educational Applications of GenAI – integrating AI into teaching, assessment, and higher education"; "Technical Foundations & Innovations – development of advanced models (LLMs, ChatGPT, GANs, NLP)" & "Ethical & Societal Dimensions – addressing issues like bias, fairness, and responsible usage".

Cluster 1 (Blue-Green Segment): This cluster includes terms such as "Generative Artificial Intelligence (GenAI), Educational Technology, Teaching, Assessment, and Higher Education", indicating a strong research focus on the integration of GenAI into learning environments and academic practices, which also shows the diffusion of GenAI into pedagogy and professional learning.

Cluster 2 (Purple Segment): Contains keywords like "Machine Learning, Large Language Models (LLMs), ChatGPT, Deep Learning, and Natural Language Processing (NLP)", highlighting the technological and methodological foundation of GenAI research and the dominance of architecture-based research.

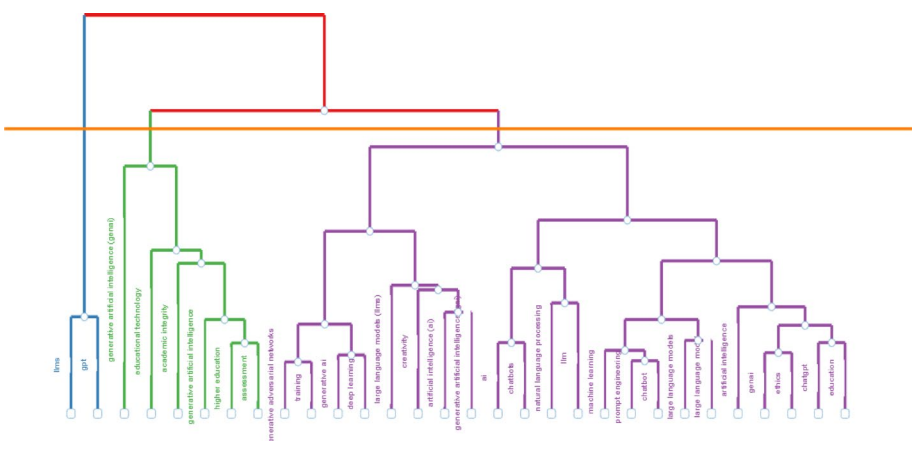

Fig. 6 Hierarchical Clustering of Keywords (Dendrogram Analysis)

Cluster 3 (Rightmost Nodes): Terms like "Ethics, AI Bias, and Ethical Technology" suggest a thematic focus on responsible AI development, ethical challenges, and societal implications, reflecting growing interest in regulatory and social dimensions.

This hierarchical clustering underscores the interdisciplinary nature of GenAI research, combining computer science, education, and ethics to advance both theoretical and applied perspectives. It also indicates that Generative AI research is evolving simultaneously along technical, ethical, and educational axes.

4.9 Factorial analysis of conceptual structure map (MCA)

Based on the MCA of keywords in Fig. 7, we can discern several thematic clusters and their relationships within the research landscape of AI, particularly concerning Generative AI, and their applications. The map reveals five clusters, each characterized by a set of highly associated keywords, visualized by polygons. Dimension 1 (Dim 1) explains 36.03% of the variance, while Dimension 2 (Dim 2) explains 26.93%. This suggests that Dim 1 captures a larger proportion of the thematic variation within the dataset. Here's a breakdown of the clusters and their interpretation:

- Cluster 1 (Blue Cluster - Upper Left Quadrant): This cluster is strongly defined by terms such as "LLM," "natural language processing," "large language models," "machine learning," "artificial intelligence," "chatbots," "ai," "genai," and "chatgpt". This cluster represents the core of general Artificial Intelligence and Large Language Models, focusing on the foundational technologies and architectures. The presence of both "Artificial Intelligence" and "Machine Learning" points to studies focusing on the basics of AI, while the presence of terms such as "chatbots" and "chatgpt" reveals more applied studies related to applications of AI in human-computer interaction.
- Cluster 2 (Red Cluster - Lower Left Quadrant): This cluster is characterized by "generative ai," "generative adversarial networks," "creativity", and "large language models". This cluster appears to represent research on the applications of Generative AI and Generative Adversarial Networks (GANs) on creativity.
- Cluster 3 (Green Cluster - Lower Right Quadrant): This cluster contains the terms "educational technology," "assigned education," "academic integrity," and "generative artificial intelligence." The "Educational Technology" cluster (Green) demonstrates

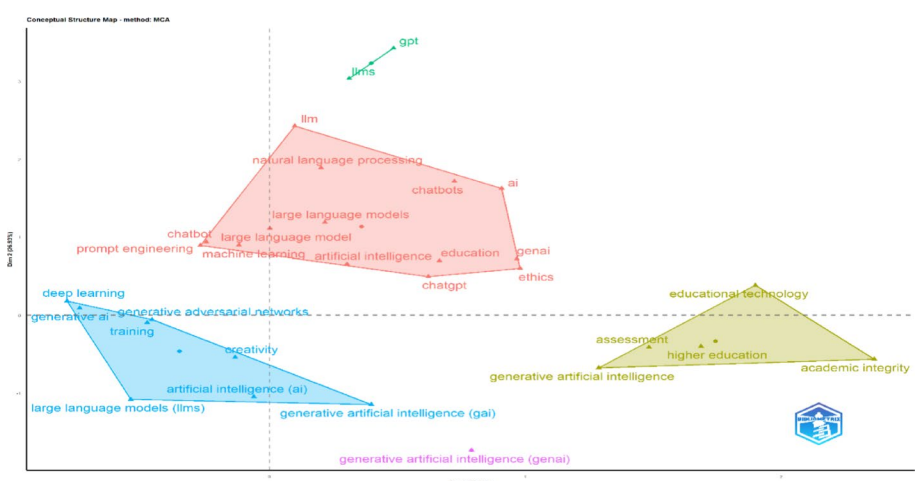

Fig. 7 Factorial Map

the growing interest in applying AI tools in education, particularly in improving academic integrity.

- Cluster 4 (Orange point - Lower Center Quadrant): This cluster is the smallest and only contains the term "generative artificial intelligence (genai)". Considering the position of this term in the graph, the cluster indicates a standalone theme of Generative AI applications that aren't explicitly linked to the other clusters.
- Cluster 5 (Purple cluster - Top Center Quadrant): The "llms" and "gpt" terms are located in this cluster. These terms represent the most recent trend in the AI field, focusing on applications related to Large Language Models.

The proximity of the "General AI and LLMs" cluster (Blue) and the "Generative AI for creativity" cluster (Red) indicates some overlap in research focus, likely concerning the use of large language models in the creation of text, images, or other media. The presence of "llms" and "gpt" as a single cluster suggests a specific, focused area of research related to newer, more powerful language models.

4.10 Latent dirichlet allocation (LDA)

The Latent Dirichlet Allocation (LDA) model identified primary topic clusters within the Generative AI literature. The evolution of topics over time revealed that early research (1986–2015) focused primarily on foundational deep learning and probabilistic models. Between 2016 and 2020, attention shifted toward applications of generative adversarial networks (GANs) and reinforcement learning. The period from 2021 onward marked a transition to large-scale LLMs, creative generation, and policy-oriented discussions surrounding ethics and regulation.

The LDA-based analysis of paper abstracts reveals ten distinct yet interrelated themes (Fig. 8) shaping the Generative AI (GenAI) research landscape. Technical streams center on system and framework design (Topic 0), language modeling and text generation (Topic 7), tool evaluations and software development (Topic 3), and image synthesis with synthetic data generation (Topic 4), highlighting engineering and methodological advancements. Complementing these are application-driven clusters, including the integration of GenAI into education for personalized learning and teaching support (Topic 2), healthcare applications in diagnostics, treatment planning, and mental health (Topic 8), as well as creative domains emphasizing content generation and human–AI collaboration (Topic 5). At a broader socio-ethical level, research interrogates governance, risks,

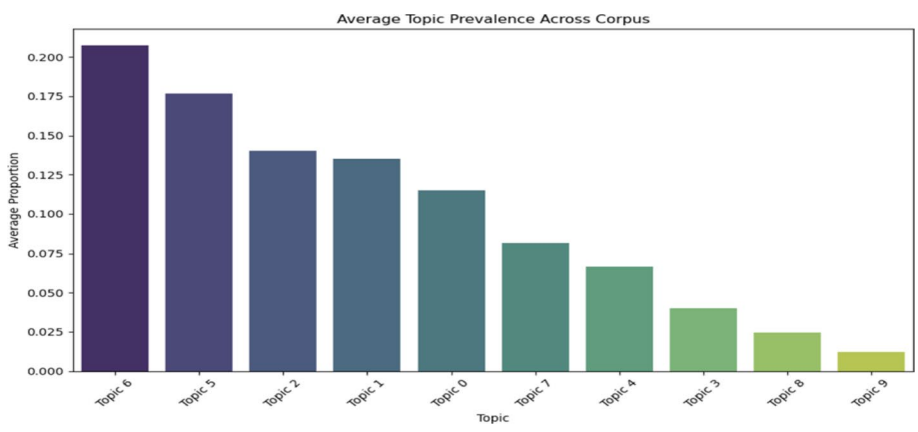

Fig. 8 Average Topic Prevalence Across Corpus

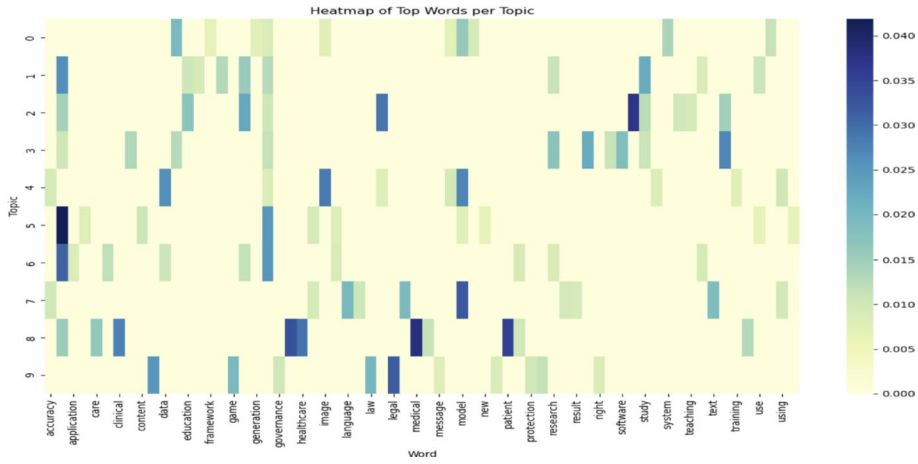

Fig. 9 Heatmap: Visual representation of topic-word strength

and responsible adoption (Topic 6), maps technological evolution and research trajectories through scientometric studies (Topic 1), and addresses legal and regulatory challenges concerning copyright, patents, and ownership of AI-generated content (Topic 9). Collectively, these themes underscore the multidisciplinary and evolving character of GenAI scholarship, spanning foundational system design, application-specific deployments, and critical reflections on policy and governance.

The heatmap of top words across the ten LDA-derived topics (Fig. 9) highlights clear thematic alignments within Generative AI (GenAI) scholarship, with governance, ethics, and implementation (Topic 6), creative content generation (Topic 5), education (Topic 2), healthcare (Topic 8), and legal–policy concerns (Topic 9) each exhibiting distinctive semantic markers, while technical themes such as system design (Topic 0), software reviews (Topic 3), image synthesis (Topic 4), and language models (Topic 7) demonstrate methodological depth. Complementing this, the topic prevalence analysis shows that strategic and governance-oriented discussions (Topic 6) dominate with over 20% of corpus weight, followed by strong emphases on content generation (≈ 17%) and educational applications (≈ 14%), reflecting growing scholarly attention to ethical adoption, creative collaboration, and pedagogical innovation. Mid-range prevalence of scientometric mapping (Topic 1, ≈ 12%) and technical system design (Topic 0, ≈ 11%) indicates balanced interest in documenting technological evolution and advancing frameworks, while lower-prevalence but high-impact niches—including software benchmarking, healthcare, image synthesis, and legal governance (≈ 2–7%)—underscore focused contributions requiring domain expertise. Collectively, these distributions reveal a GenAI research ecosystem that is both integrative and multidisciplinary, dominated by broad governance and application-oriented debates yet reinforced by specialized technical and sectoral investigations.

5 Discussion and implications

The findings reveal a rapidly expanding research landscape in Generative AI (GenAI), with a steep growth trajectory since 2022. The United States, China, India, and the United Kingdom dominate global research output, international collaborations, and citation impact, underscoring their technological and academic leadership. This dominance is strongly associated with robust funding ecosystems and institutional capacities,

as countries with well-developed research support frameworks [82–85] tend to produce more internationally collaborative and highly citable outputs. In contrast, underfunded regions often face barriers to research dissemination and global visibility. Policymakers in developing economies can leverage these insights to strengthen AI research infrastructure and foster international partnerships to mitigate geographical disparities in scholarly contributions.

Prominent institutions such as Nanyang Technological University, Harvard University, and the University of California system have emerged as leading centers driving high-impact research. Keyword analyses and thematic maps identify three dominant research dimensions viz. technical foundations (e.g., LLMs, GANs, NLP), educational applications (e.g., academic integrity, teaching, assessment), and ethical implications (e.g., responsible and transparent AI use). These results demonstrate that GenAI research is inherently interdisciplinary and globally distributed, driven by collaborative networks and continuous technological advancement.

The prominence of ethical and educational themes within emerging clusters signifies a growing awareness of responsible AI practices. Research funding agencies and institutional review boards can use these insights to design policies that promote transparency, inclusivity, and human-centered AI development. Academically, the findings suggest a shift from model-centric approaches toward broader explorations of societal, pedagogical, and governance dimensions of GenAI.

While the global dominance of the United States and China and the rise of ChatGPT-related research reflect established patterns, this study advances existing understanding by situating these developments within a broader thematic and ethical framework. The integration of topic modeling with scientometric mapping revealed emerging interdisciplinary linkages particularly those connecting technology, ethics, and education that have been underexplored in prior research. These findings indicate that GenAI research is transitioning from a purely technical phase toward a socio-technical paradigm, reflecting its deeper integration into social and institutional contexts.

Differences in citation performance across countries and institutions can be attributed to disparities in funding structures, technological infrastructure, and policy-driven incentives. Nations with established AI strategies, such as the United States, China, and members of the European Union, have made sustained strategic investments enabling large-scale projects and interdisciplinary collaborations that yield high-impact outputs. Conversely, developing economies, despite increasing publication activity, often face lower citation visibility due to limited funding, language barriers, and restricted access to high-impact journals. These findings are consistent with earlier scientometric studies [18, 86], which linked research impact to national R&D expenditure and institutional support mechanisms.

The exponential growth in GenAI publications since 2022 reflects the transformative influence of large language models (LLMs) such as GPT-3 and GPT-4 and the diffusion of generative technologies across academic and industrial domains. This surge marks a paradigm shift from algorithmic optimization and model efficiency to interdisciplinary explorations involving creativity, ethics, and human–machine collaboration. Keyword co-occurrence and LDA analyses highlight a dual focus within GenAI research advancing the technical architectures of generative models and addressing the ethical governance needed for responsible deployment. These findings align with recent studies by

Jena et al. [21] and Maheshwari et al. [20], which emphasize the ethical imperatives surrounding emerging technologies, and with earlier bibliometric analyses [18, 86], which observed the growing ethical and educational dimensions of AI.

By combining bibliometric and topic modeling techniques, this study provides a more comprehensive understanding of the structural and thematic evolution of GenAI research than traditional co-authorship or citation-based analyses. Three key implications emerge:

- **Evolution of the field**: GenAI has evolved from a niche subfield of deep learning into a multidisciplinary domain integrating ethics, creativity, and education.
- **Collaboration dynamics**: The concentration of research among select nations and institutions underscores the need for greater inclusivity and international participation.
- **Thematic convergence**: The growing overlap between technological innovation and social responsibility highlights a shift toward a balanced, human-centered AI paradigm that will shape the future trajectory of GenAI research.

Finally, the analysis reaffirms the critical role of research funding ecosystems in shaping the direction and visibility of GenAI scholarship. Countries with established AI strategies and consistent funding mechanisms such as the United States, China, and the European Union have sustained long-term research agendas and attracted global collaborations. This underscores that disparities in research influence are driven not only by publication volume but also by systemic factors such as policy continuity, cross-sector engagement, and sustained investment in AI infrastructure.

6 Limitations

Despite its comprehensive scope, this study has several limitations that should be acknowledged. This study is limited by its reliance on Scopus and Web of Science databases, which may exclude relevant research from other repositories, preprints, and non-english publications. Future research could expand coverage to include open-access repositories such as arXiv or Dimensions to capture preprints and non-English contributions. Second, the analysis was limited to publications upto 26th Aug 2025 (data taken from databases - Scopus: 1986–2025 & WoS: 2017–2025), and therefore may not fully reflect the rapidly growing literature emerging after 26th Aug. 2025. Additionally, citation-based indicators tend to favor older publications, potentially underestimating the impact of recent high-quality work. Future research could incorporate alternative metrics (e.g., Altmetrics) to capture online engagement and broader societal impact. Longitudinal analyses may also be conducted to track evolving research themes and identify disruptive innovations in GenAI. Also, while Biblioshiny and VOSviewer provide robust tools for bibliometric visualization, certain interpretive nuances may have been influenced by keyword normalization and database indexing inconsistencies.

Another important limitation of this study lies in the search strategy used to retrieve bibliographic data. The dataset was collected using the keyword "Generative AI" and its variants ("gen AI," "genai") to maintain precision and focus. However, related terms such as "large language models," "LLMs," "ChatGPT," and "transformer models" were not explicitly included in the search query. Consequently, some publications that discuss Generative AI concepts under these specific terminologies might not have been

captured in the dataset. Future research could broaden the scope of keyword selection to include these emerging terms in order to provide a more comprehensive representation of the GenAI research landscape.

Additionally, the Latent Dirichlet Allocation (LDA) topic modeling approach, while effective in identifying thematic clusters, relies on unigram tokenization, which may simplify complex relationships between research concepts. Future studies could adopt bi-gram or contextual topic modeling to enhance interpretive depth. Finally, this study focused primarily on structural and thematic analysis; integrating qualitative content analysis or citation context studies could provide richer insights into the conceptual evolution of Generative AI research.

7 Conclusion

This study provides a detailed overview of growth and intellectual structure of GenAI research. The analysis reveals a remarkable surge in scholarly activity since 2022, with exponential expansion during 2023–2024, underscoring GenAI's rapid transition from a niche innovation to a transformative, general-purpose technology. Research production, though globally distributed, remains concentrated in technologically advanced nations such as the United States, China, India, and the United Kingdom, supported by eminent institutions including Nanyang Technological University, Harvard University, and the University of California System.

The findings demonstrate the distinctly interdisciplinary nature of GenAI research, integrating technical, social, and ethical dimensions. Core themes encompass foundational models (LLMs, GANs, diffusion models), educational innovations, healthcare applications, and pressing debates surrounding ethics, governance, and intellectual property. The Latent Dirichlet Allocation (LDA) topic modeling further indicates that concerns about ethical governance, responsible implementation, and risk management dominate current discourse, reflecting an academic shift toward socially conscious and sustainable AI development. At the same time, the growing focus on creativity, pedagogy, and healthcare highlights the translational and cross-sectoral potential of GenAI technologies.

Methodologically, this study advances scientometric research by integrating bibliometric mapping with topic modeling, thereby uncovering the conceptual, thematic, and collaborative dynamics shaping the GenAI landscape. The use of network visualizations and scientometric indicators provides a transparent and replicable analytical framework.

The study's implications extend across multiple domains. For scholars, it identifies evolving research trajectories, intellectual linkages, and knowledge gaps that can guide future inquiry. For practitioners and industry stakeholders, it highlights cutting-edge innovations and collaborative networks that can inform the responsible integration of GenAI into real-world systems. For policymakers, it offers data-driven evidence to support informed decision-making, strategic resource allocation, and the formulation of ethical governance frameworks that foster sustainable AI ecosystems.

Beyond descriptive mapping, this study contributes interpretive depth by connecting publication and thematic trends with the broader ethical and institutional contexts of AI development. By revealing how research on GenAI increasingly intertwines technological progress with moral responsibility and human-centered values, the study underscores the need for continued balance between innovation and ethical stewardship.

As GenAI continues to evolve, sustained scientometric monitoring and interdisciplinary collaboration will be vital to navigate its expanding frontiers. The insights from this study can guide researchers, educators, and policymakers toward fostering an AI ecosystem that is not only technologically advanced but also ethically grounded, socially equitable, and globally inclusive. In doing so, GenAI can truly fulfill its potential as a transformative force for knowledge creation, innovation, and societal progress.

Author contributions
KP & RV conceived the idea. RV downloaded the dataset. KP wrote Introduction and literature review and LDA. RV and AK prepared analysis and wrote Discussion, Limitations, along with the conclusion. All authors reviewed the manuscript.

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Data availability
All data generated and analysed during this study are available with the authors and shall be made available if required.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Clinical trial
Not applicable.

AI-assisted writing disclosure
The conception of the research idea, methodological design, data analysis, and Python code implementation were entirely carried out by the authors. Parts of the text were refined using ChatGPT (OpenAI) to enhance readability and clarity; however, all analyses, interpretations, and conclusions were solely designed, verified, and approved by the authors.

Competing interests
The authors declare no competing interests.

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